

Analytical Prediction of Stability Lobes in Milling

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Abstract

A new method for the analytical prediction of stability lobes in milling is presented. The stability model requires transfer functions of the structure at the cutter - workpiece contact zone, static cutting force coefficients, radial immersion and the number of teeth on the cutter. Time varying dynamic cutting force coefficients are approximated by their Fourier series components, and the chatter free axial depth of cuts and spindle speeds are calculated directly from the proposed set of linear analytic expressions without any digital iteration. Analytically predicted stability lobes are compared with the lobes generated by time domain and other numerical methods available in the literature.

Keywords: Chatter, stability, milling.

1 Introduction

The machine tool chatter vibrations occur due to a self excitation mechanism in the generation of chip thickness during machining operations. In milling, one of the structural modes of the machine tool - workpiece system is excited by cutting forces initially. An oscillatory surface finish left by one of the tooth is removed by the succeeding oscillatory tooth due to structural vibrations, see figure 1. The resulting chip thickness becomes also oscillatory, which in turn produces oscillatory cutting forces whose magnitudes are proportional to the time varying chip load. The self excited cutting system becomes unstable, and chatter vibrations grow until the tool jumps out of the cut or breaks under the excessive cutting forces. Thus, the chatter vibrations continue to be the major limiting factor in increasing the metal removal rates of the machine tools.

Tobias [1] and Tlusty [2] presented basics of chatter vibrations on machine tools by explaining the fundamental mechanism of regeneration of chip thickness. Tlusty derived the fundamental stability theory in orthogonal cutting where the chatter vibrations free axial depth of cut is shown to be inversely proportional to cutting coefficients and the amplitude of the real part of the transfer function between the tool and workpiece. Tobias [1] also presented a similar stability law, and added process stiffness and damping terms produced by the dynamic motion of the tool during cutting. By analyzing the phase shift between the inner and outer modulations on the wavy chip, Tobias [1] presented a method of generating stability lobes which contain chatter vibration free depth of cuts

and cutting speeds.

The analytical prediction of stability lobes presented by Tlusty [2], Tobias [1] and Merrit [3] are mostly applicable to operations like turning, which resembles the most basic, orthogonal cutting system where the direction of the cutting forces and chip thickness generation are time invariant. In the case of milling however, the chip thickness, the cutting forces and the direction of excitation vary and are intermittent due to rotating cutter. Knowing the difficulty in the realistic application of basic stability law to the process, Tlusty et al. [4, 5] presented a method of generating stability lobes using time domain simulations of the chatter vibrations in milling. They considered the velocity dependent process damping and the basic nonlinearity in the chatter which arises when the tool jumps out of the cut due to excessive vibrations. Tlusty and Smith [13], and Weck and Altıntaş [14] incorporated the simulated stability lobes to a CAD/CAM system for chatter free process planning of NC tool paths. Altıntaş et al. [6] incorporated the process damping and stiffness to the time domain simulation model, and predicted the resulting surface finish produced by chatter vibrations in milling. Recently, Heisel [7] presented a method of identifying chatter vibration characteristics from surface measurements.

Although time domain numerical simulation methods are quite powerful since they allow the nonlinearities and various tool geometries, a pure analytical method is still desirable to understand the mechanism, and for rapid and easy prediction of chatter free cutting conditions in milling. The first detailed mathematical model of the dynamic milling process was derived by Sridhar et al. [8].

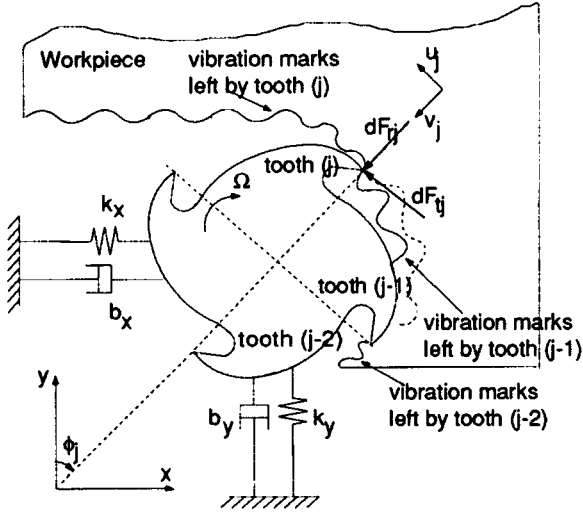


Figure 1: Dynamic model of milling with two degrees of freedom

They have considered the time varying directional cutting force coefficients in deriving the dynamic regenerative milling forces, and solved the chatter stability by a numerical method [8]. Recently, Minis and Yanushevsky [9] presented a comprehensive analytical method in solving the dynamic milling model presented by Sridhar et al. [8]. They have applied the available theory of periodic differential - difference equations to dynamic milling. The critical axial depth of cut was predicted by an iterative checking of an analytic stability expression.

In this paper, the stability of chatter in milling is solved by an alternative method, which is more practical and resembles the basic chatter stability laws of Tobias [1] and Tlustý [2] developed for orthogonal cutting. The dynamic milling process model is derived by considering the Fourier series expansion of the time varying milling force coefficients. The eigenvalues of the dynamic milling expression are calculated analytically by selecting a chatter frequency around dominant structural modes. Noting a physical fact that the axial depth of cut is always a real quantity, the chatter free axial depth of cuts and spindle speeds are analytically formulated as a function of structure's transfer functions at the tool - workpiece contact zone, static cutting force constants, number of teeth and radial depth of cut. The analytical expressions provide the same results obtained by the numerical solution of Sridhar [8], iterative analytical solution of Minis et al [9] and time domain simulations of Smith and Tlustý [12].

2 Dynamic Milling Model

Milling cutters can be considered to have two orthogonal degrees of freedom as shown in figure 1. The cutter is assumed to have N number of teeth with a zero helix angle. The cutting forces

excite the structure in the feed (X) and normal (Y) directions, causing dynamic displacements x and y respectively. The dynamic displacements are carried to rotating tooth number (j) in the radial or chip thickness direction with the coordinate transformation of $v_j = -x \sin \phi_j - y \cos \phi_j$ where ϕ_j is the instantaneous angular immersion of tooth (j) measured clockwise from the normal (Y) axis. If the spindle rotates at an angular speed of Ω (rad/s), the immersion angle vary with time as $\phi_j(t) = \Omega t$. The resulting chip thickness consists of static part ($s_t \sin \phi_j$), which is due to rigid body motion of the cutter, and dynamic component caused by the vibrations of the tool at the present and previous tooth periods. Since the chip thickness is measured in the radial direction (v_j), the total chip load can be expressed by,

$$h(\phi_j) = [s_t \sin \phi_j + (v_{j,0} - v_j)]g(\phi_j) \quad (1)$$

where s_t is the feed rate per tooth and $(v_{j,0}, v_j)$ are the dynamic displacements of the cutter at the previous and present tooth periods, respectively. $g(\phi_j)$ is a unit step function which determines whether the tooth is in or out of cut, i.e.,

$$\left. \begin{aligned} g(\phi_j) &= 1 \leftarrow \phi_{st} < \phi_j < \phi_{ex} \\ g(\phi_j) &= 0 \leftarrow \phi_j < \phi_{st} \text{ or } \phi_j > \phi_{ex} \end{aligned} \right\} \quad (2)$$

where ϕ_{st}, ϕ_{ex} are start and exit immersion angles of the cutter to and from the cut, respectively. Henceforth, the static component of the chip thickness ($s_t \sin \phi_j$) is dropped from the expressions since it does not contribute to the dynamic chip load regeneration mechanism. Substituting v_j into (1) yields,

$$h(\phi_j) = [\Delta x \sin \phi_j + \Delta y \cos \phi_j]g(\phi_j) \quad (3)$$

where $\Delta x = x - x_0, \Delta y = y - y_0$. (x, y) and (x_0, y_0) represent the dynamic displacements of the cutter structure at the present and previous tooth periods, respectively. The tangential (F_{tj}) and radial (F_{rj}) cutting forces acting on the tooth j is proportional to the axial depth of cut (a) and chip thickness (h),

$$F_{tj} = K_t a h(\phi_j), \quad F_{rj} = K_r F_{tj} \quad (4)$$

where cutting coefficients K_t and K_r are constant. Resolving the cutting forces in the x and y directions,

$$\begin{aligned} F_{xj} &= -F_{tj} \cos \phi_j - F_{rj} \sin \phi_j \\ F_{yj} &= +F_{tj} \sin \phi_j - F_{rj} \cos \phi_j \end{aligned} \quad (5)$$

and summing the cutting forces contributed by all teeth, the total dynamic milling forces acting on the cutter are found as,

$$F_x = \sum_{j=0}^{N-1} F_{xj} ; \quad F_y = \sum_{j=0}^{N-1} F_{yj} \quad (6)$$

where $\phi_j = \phi_p + j\phi$, and cutter pitch angle is $\phi_p = 2\pi/N$. Substituting the chip thickness (3) and tooth forces (4) into (5), and rearranging the resulting expressions in matrix form yields,

$$\begin{Bmatrix} F_x \\ F_y \end{Bmatrix} = \frac{1}{2} a K_t \begin{bmatrix} a_{xx} & a_{xy} \\ a_{yx} & a_{yy} \end{bmatrix} \begin{Bmatrix} \Delta x \\ \Delta y \end{Bmatrix} \quad (7)$$

where time varying directional dynamic milling force coefficients are given by,

$$\begin{aligned} a_{xx} &= \sum_{j=0}^{N-1} -g_j [\sin 2\phi_j + K_r (1 - \cos 2\phi_j)] \\ a_{xy} &= \sum_{j=0}^{N-1} -g_j [(1 + \cos 2\phi_j) + K_r \sin 2\phi_j] \\ a_{yx} &= \sum_{j=0}^{N-1} g_j [(1 - \cos 2\phi_j) - K_r \sin 2\phi_j] \\ a_{yy} &= \sum_{j=0}^{N-1} g_j [\sin 2\phi_j - K_r (1 + \cos 2\phi_j)]. \end{aligned}$$

Considering that the angular position of the parameters change with time and angular velocity, the equation (7) can be expressed in time domain in a matrix form as,

$$\{F(t)\} = \frac{1}{2} a K_t [A(t)] \{\Delta(t)\} \quad (8)$$

As cutter rotates, the directional factors vary with time, which is the fundamental difference between milling and operations like turning where the direction of the force is constant. However, like the milling forces, $[A(t)]$ is periodic at tooth passing frequency $\omega = N\Omega$ or tooth period $T = 2\pi/\omega$, thus can be expanded into Fourier series.

$$\begin{aligned} [A(t)] &= \sum_{r=-\infty}^{\infty} [A_r] e^{ir\omega t} \\ [A_r] &= \frac{1}{T} \int_0^T [A(t)] e^{-ir\omega t} dt \end{aligned} \quad (9)$$

The number of harmonics (r) of the tooth passing frequency (ω) to be considered for an accurate reconstruction of $[A(t)]$ depends on the immersion conditions and the number of teeth in cut. If most simplistic approximation, the average component of the Fourier series expansion, is considered, i.e. $r = 0$,

$$[A_0] = \frac{1}{T} \int_0^T [A(t)] dt. \quad (10)$$

Since $[A_0]$ is valid only between the entry (ϕ_{st}) and exit (ϕ_{ex}) angles of the cutter (i.e. $g_j(\phi_j) = 1$), it becomes equal to the average value of $[A(t)]$ at cutter pitch angle $\phi_p = 2\pi/N$.

$$[A(0)] = \frac{1}{\phi_p} \int_{\phi_{st}}^{\phi_{ex}} [A(\phi)] d\phi = \frac{N}{2\pi} \begin{bmatrix} \alpha_{xx} & \alpha_{xy} \\ \alpha_{yx} & \alpha_{yy} \end{bmatrix} \quad (11)$$

where the integrated functions are given as:

$$\begin{aligned} \alpha_{xx} &= \frac{1}{2} \left[\cos 2\phi - 2K_r \phi + K_r \sin 2\phi \right]_{\phi_{st}}^{\phi_{ex}} \\ \alpha_{xy} &= \frac{1}{2} \left[-\sin 2\phi - 2\phi + K_r \cos 2\phi \right]_{\phi_{st}}^{\phi_{ex}} \\ \alpha_{yx} &= \frac{1}{2} \left[-\sin 2\phi + 2\phi + K_r \cos 2\phi \right]_{\phi_{st}}^{\phi_{ex}} \\ \alpha_{yy} &= \frac{1}{2} \left[-\cos 2\phi - 2K_r \phi - K_r \sin 2\phi \right]_{\phi_{st}}^{\phi_{ex}} \end{aligned}$$

The dynamic milling expression (8) is therefore reduced to the following

$$\{F(t)\} = \frac{1}{2} a K_t [A_0] \{\Delta(t)\} \quad (12)$$

where $[A_0]$ is time invariant but immersion dependent directional cutting coefficient matrix. Since the average cutting force per tooth period is independent of the helix angle, $[A_0]$ is valid for helical end mills as well.

3 Chatter Stability Lobes

Transfer function matrix ($[G(i\omega)]$) identified at the cutter - workpiece contact zone,

$$[G(i\omega)] = \begin{bmatrix} G_{xx}(i\omega) & G_{xy}(i\omega) \\ G_{yx}(i\omega) & G_{yy}(i\omega) \end{bmatrix} \quad (13)$$

where $G_{xx}(i\omega)$ and $G_{yy}(i\omega)$ are the direct transfer functions in x and y directions, and $G_{xy}(i\omega)$ and $G_{yx}(i\omega)$ are the cross transfer functions. The vibration vectors at the present time (t) and previous tooth period ($t - T$) are defined as,

$$\{r\} = \{x(t) \ y(t)\}^T; \quad \{r_0\} = \{x(t-T) \ y(t-T)\}^T$$

Describing the vibrations at the chatter frequency ω_c in the frequency domain using harmonic functions,

$$\begin{aligned} \{r(i\omega_c)\} &= [G(i\omega)] \{F\} e^{i\omega_c t} \\ \{r_0(i\omega_c)\} &= e^{-i\omega_c T} \{r(i\omega_c)\} \end{aligned} \quad (14)$$

and substituting $\{\Delta\} = \{(x - x_0) \ (y - y_0)\}^T$ gives,

$$\begin{aligned} \{\Delta(i\omega_c)\} &= \{r(i\omega_c)\} - \{r_0(i\omega_c)\} \\ &= [1 - e^{-i\omega_c T}] e^{i\omega_c t} [G(i\omega_c)] \{F\} \end{aligned}$$

where $\omega_c T$ is the phase delay between the vibrations at successive tooth periods T . Substituting $\{\Delta(i\omega_c)\}$ into the dynamic milling equation (12) gives,

$$\{F\} e^{i\omega_c t} = \frac{1}{2} a K_t [1 - e^{-i\omega_c T}] [A_0] [G(i\omega_c)] \{F\} e^{i\omega_c t}$$

which has a non-trivial solution if its determinant is zero,

$$\det[[I] - \frac{1}{2}K_t a(1 - e^{-i\omega_c T})[A_0][G(i\omega_c)]] = 0$$

which is the characteristic equation of the closed loop dynamic milling system. The notation is further simplified by defining the oriented transfer function matrix as,

$$[G_0(i\omega_c)] = \begin{bmatrix} \alpha_{xx}G_{xx}(i\omega_c) + \alpha_{xy}G_{yx}(i\omega_c) & \alpha_{xx}G_{xy}(i\omega_c) + \alpha_{xy}G_{yy}(i\omega_c) \\ \alpha_{yx}G_{xx}(i\omega_c) + \alpha_{yy}G_{yx}(i\omega_c) & \alpha_{yx}G_{xy}(i\omega_c) + \alpha_{yy}G_{yy}(i\omega_c) \end{bmatrix} \quad (15)$$

and the eigenvalue of the characteristic equation as,

$$\Lambda = -\frac{N}{4\pi}aK_t(1 - e^{-i\omega_c T}). \quad (16)$$

The resulting characteristic equation becomes,

$$\det[[I] + \Lambda[G_0(i\omega_c)]] = 0. \quad (17)$$

The eigenvalue of the above equation can easily be solved for a given chatter frequency ω_c , static cutting factors (K_t, K_r) which can be stored as a material dependent quantity for any milling cutter geometry [11], radial immersion (ϕ_{st}, ϕ_{ex}) and transfer function of the structure (15). If two orthogonal degrees of freedom in feed (X) and normal (Y) directions are considered (i.e. $G_{xy} = G_{yx} = 0.0$), the characteristic equation becomes just a quadratic function

$$a_0\Lambda^2 + a_1\Lambda + 1 = 0 \quad (18)$$

where

$$\begin{aligned} a_0 &= G_{xx}(i\omega_c)G_{yy}(i\omega_c)(\alpha_{xx}\alpha_{yy} - \alpha_{xy}\alpha_{yx}) \\ a_1 &= \alpha_{xx}G_{xx}(i\omega_c) + \alpha_{yy}G_{yy}(i\omega_c) \end{aligned}$$

Then, the eigenvalue Λ is obtained as:

$$\Lambda = -\frac{1}{2a_0}(a_1 \pm \sqrt{a_1^2 - 4a_0}). \quad (19)$$

As long as the plane of cut (x, y) is considered, the characteristic equation is still a simple quadratic function regardless of the number of modes considered in the machine tool structure. Since the transfer functions are complex, the eigenvalue has a real and an imaginary part, $\Lambda = \Lambda_R + i\Lambda_I$. Substituting the eigenvalue and $e^{-i\omega_c T} = \cos \omega_c T - i \sin \omega_c T$ in equation (16) gives the critical axial depth of cut at chatter frequency ω_c ,

$$\begin{aligned} a_{lim} = & -\frac{2\pi}{N K_t} \left[\frac{\Lambda_R(1 - \cos \omega_c T) + \Lambda_I \sin \omega_c T}{(1 - \cos \omega_c T)} \right. \\ & \left. + i \frac{\Lambda_I(1 - \cos \omega_c T) - \Lambda_R \sin \omega_c T}{(1 - \cos \omega_c T)} \right] \quad (20) \end{aligned}$$

Since a_{lim} is a real number, the imaginary part of the equation (20) must vanish,

$$\Lambda_I(1 - \cos \omega_c T) - \Lambda_R \sin \omega_c T = 0 \quad (21)$$

By substituting,

$$\kappa = \frac{\Lambda_I}{\Lambda_R} = \frac{\sin \omega_c T}{1 - \cos \omega_c T}. \quad (22)$$

into the real part of the equation (20) (imaginary part vanishes), the final expression for chatter free axial depth of cut is found as:

$$a_{lim} = -\frac{2\pi\Lambda_R}{N K_t}(1 + \kappa^2) \quad (23)$$

Therefore, given the chatter frequency (ω_c), the chatter limit in terms of the axial depth of cut can directly be determined from equation (23), which has a similar form with the classical orthogonal cutting stability model proposed by Tlusty [2]. The corresponding spindle speeds are also found in a trivial manner. From equation (22), the angular distance travelled by the tooth due to chatter frequency ω_c at tooth period T is found as

$$\omega_c T = \cos^{-1} \frac{\kappa^2 - 1}{\kappa^2 + 1} = -\cos^{-1} 2\psi. \quad (24)$$

Note that $\kappa = \Lambda_I/\Lambda_R = \tan \psi$ and ψ is the phase shift of the eigenvalue. Thus if k is the integer number of full vibration waves (i.e. lobes) imprinted on the cut arc,

$$\omega_c T = \pi - 2\psi + 2k\pi = \epsilon + 2k\pi \quad (25)$$

where $\psi = \tan^{-1} \kappa$ and $\epsilon = \pi - 2\psi$ is the phase shift between inner and outer modulations (present and previous vibration marks). (Note that if the real part of the eigenvalue $\Lambda_R < 0$ than additional π must be added to ψ in the digital calculations.) The spindle speed $n(\text{rev/min})$ is simply calculated by finding the tooth passing period $T(s)$,

$$T = \frac{1}{\omega_c}(\epsilon + 2k\pi) \rightarrow n = \frac{60}{NT} \quad (26)$$

In summary, the transfer functions of the machine tool system are identified, and the dynamic cutting coefficients are evaluated from the derived equation (11) for a specified cutter, workpiece material and radial immersion of the cut. Then the stability lobes are calculated as follows:

- select a chatter frequency from transfer functions around a dominant mode,
- solve the eigenvalue equation (18),
- calculate the critical depth of cut from equation (23),
- calculate the spindle speed from equation (26) for each stability lobe $k = 0, 1, 2, \dots$
- repeat the procedure by scanning the chatter frequencies around all dominant modes of the structure evident on the transfer functions.

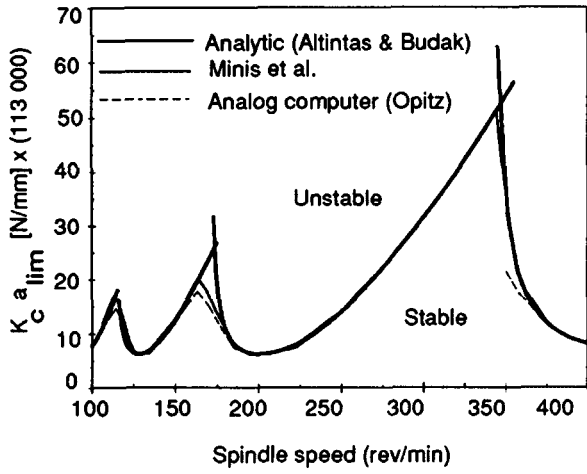


Figure 2: A single single degree of freedom system: Comparison of the predicted chatter limit with the published data (Minis and Yanushkevsky [1993], analog data is from Opitz [1968] and Sridhar et al. [1968]).

4 Applications

Milling of a SDOF Structure: The stability formula is applied to a milling operation investigated by Opitz [15], Sridhar et al. [8] and Minis et al. [10] for comparison. The flexibility of the workpiece in the x direction is represented by a single degree-of-freedom system with the following parameters: $k_x = 7.152 \times 10^6$ N/m, $\xi_x = 0.0417$, $\omega_{nx} = 355$ rad/sec. Substituting in eqn. 15 gives the oriented transfer function of this simple case,

$$G_{0x} = \alpha_{xx} \frac{\omega_{nx}^2 k_x}{\omega_{nx}^2 - \omega_c^2 + i2\xi\omega_{nx}\omega_c}$$

which yields to a simple eigenvalue of $\Lambda = 1/a_1 = 1/G_{0x}$ and the critical depth of cut:

$$a_{lim} = \frac{2\pi}{N\alpha_{xx}K_t Re[G_{xx}(i\omega_c)]}$$

The above equation for this simple milling case has an identical form to the stability law proposed by Tlustý [2] but with an immersion dependent cutting force coefficient. (α_{xx}) depends on the cutter immersion, up or down milling modes and the radial cutting constant K_r . The product of α_{xx} and the real part of the structure's transfer function $Re[G_{xx}(i\omega_c)]$ at the chatter frequency must be positive. A face milling cutter with $N = 10$ straight teeth is used, and the start and exit angles are $\phi_{st} = 67^\circ$ and $\phi_{ex} = 139^\circ$, respectively. $K_r = 0.577$ and α_{xx} is calculated from equation (11) as (-1.6). Equation (23) was used to calculate the stability limit for different values of chatter frequency ω_c . Then, the corresponding spindle speeds are found from equation (26). Stability lobes are shown in figure 2 in terms of the product

of critical depth of cut a_{lim} and the resultant cutting force coefficient, i.e. $K_c = K_t \sqrt{1 + K_r^2}$ as in [8]. Also shown in figure 2 is the data from Minis et al. [10] and Sridhar et al. [8]. Minis et al. [10] iteratively determined the stability limit for a given spindle speed by examining the stability of the system for different axial depths. They used periodic system theory in formulating the milling stability problem. Sridhar et al. [8] used the analog simulation data from Opitz [15] to compare with their numerical stability solution results. All three solutions, which use different approaches, show excellent agreement.

Milling with a 2-DOF Cutter : A two DOF milling system example is taken from Smith and Tlustý [12]. This is a half immersion-up milling of

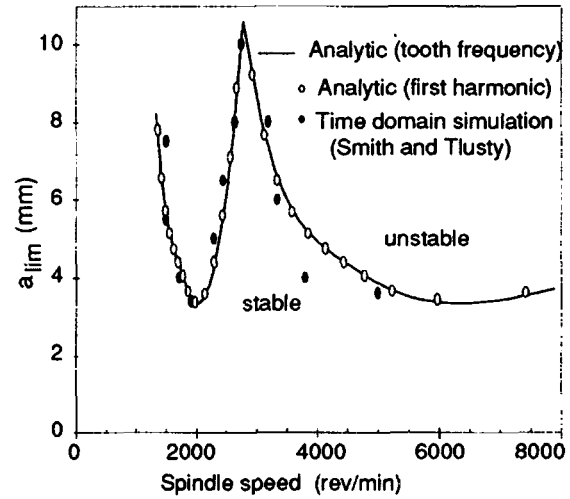


Figure 3: Analytical, multi frequency and time domain stability limit predictions for a case analyzed by Smith and Tlustý [1990].

an aluminum workpiece by a 4 in. diameter shell mill with 8 teeth. The modal properties of the cutter are given as follows:

	Mode	ω_n (Hz)	k (N/m)	ξ (%)
X	1	260	2.26×10^8	0.12
	2	389	5.54×10^7	0.04
Y	1	150	2.13×10^8	0.1
	2	348	2.14×10^7	0.1

The results of the analytical method predictions ($K_t = 1500$ MPa and $K_r = 0.3$) and the time domain simulations performed by Smith and Tlustý [12] are shown in figure 3. A multi frequency solution is also included by considering the first harmonic term ($r = 0, 1$) in the Fourier series in equation (9). The eigenvalue of this higher order equation is numerically evaluated. In the cases, where the AC component of the directional milling coefficients is higher (i.e. small number of teeth and low radial depth of cut), at least the first or higher harmonic terms in the Fourier series may be required for the accurate solutions. For

this case however, the single frequency approximation, which provides a complete analytical solution without any iteration, is in excellent agreement with the time domain simulations and multi frequency solution.

5 Conclusion

An analytical method of predicting stability limits in milling is presented. The method is based on the formulation of dynamic milling with regeneration in the chip thickness, time varying directional factors and the interaction with the machine tool structure. By scanning a range of frequencies where chatter vibrations occur, i.e. around the structural modes of the system, the chatter free axial depth of cuts and spindle speeds are directly calculated using the proposed analytical but practical expressions. The method presented here was summarized from a more general theory derived for the peripheral helical milling of very flexible thin aerospace webs, which have varying structural dynamics in the direction of axial depth of cut as well. The complete details of the multi-dimensional solution are given in the doctoral thesis of Budak [16]. The method requires further investigation for the cutting of small radial depth of cuts with helical end mills having two or three flutes.

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